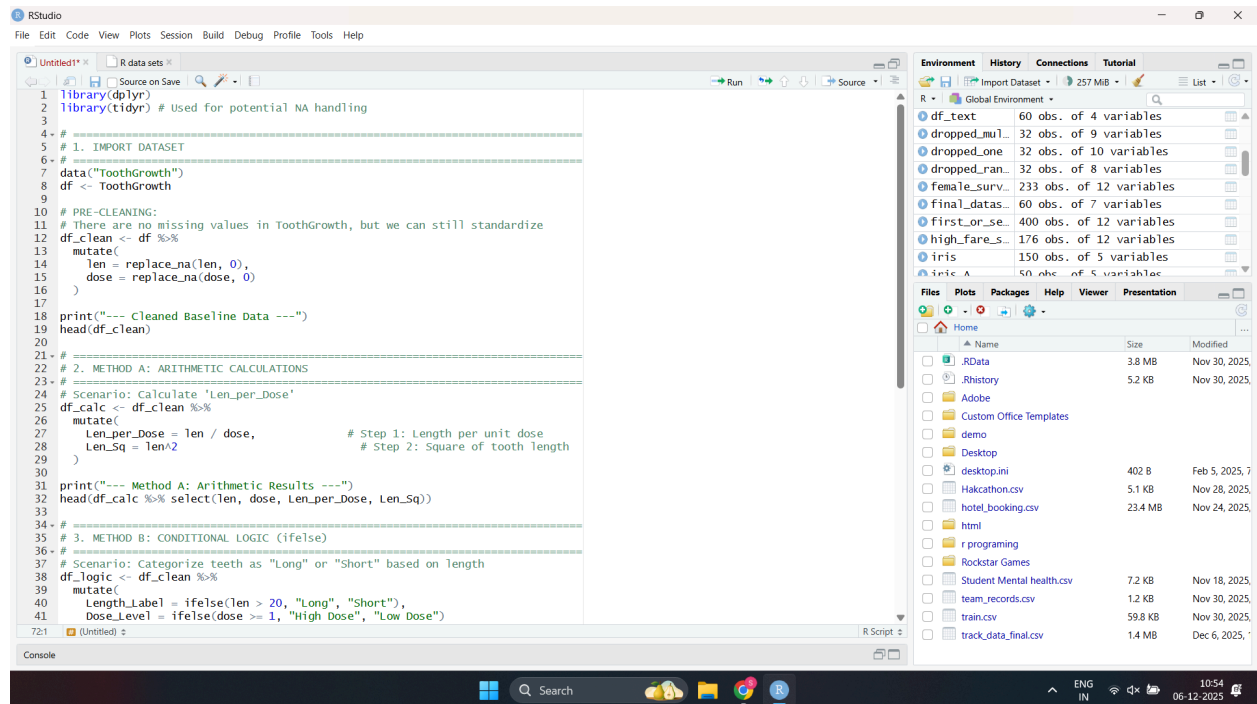


SHETH L.U.J. & SIR M.V. COLLEGE OF SCIENCE
SUBJECT - Data Analysis SAS / SPSS / R

Aim - Creating new variables using transformations and calculations in R. import dataset.

Input :



```
1 library(dplyr)
2 library(tidyrr) # Used for potential NA handling
3
4 # =====
5 # 1. IMPORT DATASET
6 # =====
7 data("ToothGrowth")
8 df <- ToothGrowth
9
10 # PRE-CLEANING:
11 # There are no missing values in ToothGrowth, but we can still standardize
12 df_clean <- df %>%
13   mutate(
14     len = replace_na(len, 0),
15     dose = replace_na(dose, 0)
16   )
17
18 print("--- Cleaned Baseline Data ---")
19 head(df_clean)
20
21 # =====
22 # 2. METHOD A: ARITHMETIC CALCULATIONS
23 # =====
24 # Scenario: Calculate 'Len_per_Dose'
25 df_calc <- df_clean %>%
26   mutate(
27     Len_per_Dose = len / dose,          # Step 1: Length per unit dose
28     Len_Sq = len^2                      # Step 2: Square of tooth length
29   )
30
31 print("--- Method A: Arithmetic Results ---")
32 head(df_calc %>% select(len, dose, Len_per_Dose, Len_Sq))
33
34 # =====
35 # 3. METHOD B: CONDITIONAL LOGIC (ifelse)
36 # =====
37 # Scenario: Categorize teeth as "Long" or "Short" based on length
38 df_logic <- df_clean %>%
39   mutate(
40     Length_Label = ifelse(len > 20, "Long", "Short"),
41     Dose_Level = ifelse(dose >= 1, "High Dose", "Low Dose")
42   )
43
44 print("--- Method B: Logic Results ---")
45 head(df_logic %>% select(len, Length_Label, dose, Dose_Level))
46
47 # =====
48 # 4. METHOD C: TEXT TRANSFORMATION (paste)
49 # =====
50 # Scenario: Create a summary column combining supplement type and length info
51 df_text <- df_clean %>%
52   mutate(
53     Tooth_Summary = paste("Supplement:", supp, "| Length:", len, "| Dose:", dose)
54   )
55
56 print("--- Method C: Text Transformation ---")
57 head(df_text %>% select(Tooth_Summary))
58
59 # =====
60 # 5. ALL TOGETHER (The Standard Workflow)
61 # =====
62 final_dataset <- df_clean %>%
63   mutate(
64     Len_per_Dose = len / dose,
65     Len_Category = ifelse(len > 20, "Long Tooth", "Short Tooth"),
66     Dose_Level = ifelse(dose >= 1, "High Dose", "Low Dose"),
67     Tooth_Report = paste0("Supplement: ", supp, ", Length: ", len, ", Dose: ", dose)
68   )
69
70 print("--- Final Combined Dataset ---")
71 head(final_dataset)
72
73
```

Output :

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Roll No - S073

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SUBJECT - Data Analysis SAS / SPSS / R

```
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Source
Console Terminal
R - R4.5.2 - ~/R/
> library(dplyr)
> library(tidyverse) # Used for potential NA handling
> #
> # 1. IMPORT DATASET
> #
> data("ToothGrowth")
> df <- ToothGrowth
> # PRE-CLEANING:
> # There are no missing values in ToothGrowth, but we can still standardize
> df_clean <- df %>%
+   mutate(
+     len = replace_na(len, 0),
+     dose = replace_na(dose, 0)
+   )
> print("--- Cleaned Baseline Data ---")
[1] "--- Cleaned Baseline Data ---"
> head(df_clean)
  len supp dose
1  4.2  VC  0.5
2 11.5  VC  0.5
3  7.3  VC  0.5
4  5.8  VC  0.5
5  6.4  VC  0.5
6 10.0  VC  0.5

> # 2. METHOD A: ARITHMETIC CALCULATIONS
> #
> # Scenario: Calculate 'len_per_dose'
> df_calc <- df_clean %>%
+   mutate(
+     len_per_dose = len / dose,          # Step 1: Length per unit dose
+     len_sq = len^2                      # Step 2: Square of tooth length
+   )
> print("--- Method A: Arithmetic Results ---")
[1] "--- Method A: Arithmetic Results ---"
> head(df_calc %>% select(len, dose, len_per_dose, len_sq))
  len dose len_per_dose len_sq
1  4.2  0.5         8.4   17.64
2 11.5  0.5        23.0  132.25
3  7.3  0.5        14.6   53.29
4  5.8  0.5        11.6   33.64
5  6.4  0.5        12.8   40.96

Environment History Connections Tutorial
R - Global Environment
df_text 60 obs. of 4 variables
dropped_mu1... 32 obs. of 9 variables
dropped_one 32 obs. of 10 variables
dropped_ran... 32 obs. of 8 variables
female_surv... 233 obs. of 12 variables
final_datas... 60 obs. of 7 variables
first_or_se... 400 obs. of 12 variables
high_fare_s... 176 obs. of 12 variables
iris 150 obs. of 5 variables
iris_A 50 obs. of 5 variables

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```

```
RStudio
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Source
Console Terminal
R - R4.5.2 - ~/R/
> # 3. METHOD B: CONDITIONAL LOGIC (ifelse)
> #
> # Scenario: Categorize teeth as "Long" or "Short" based on length
> df_logic <- df_clean %>%
+   mutate(
+     Length_Label = ifelse(len > 20, "Long", "Short"),
+     Dose_Level = ifelse(dose >= 1, "High Dose", "Low Dose")
+   )
> print("--- Method B: Logic Results ---")
[1] "--- Method B: Logic Results ---"
> head(df_logic %>% select(len, Length_Label, dose, Dose_Level))
  len Length_Label dose Dose_Level
1  4.2      Short  0.5   Low Dose
2 11.5      Short  0.5   Low Dose
3  7.3      Short  0.5   Low Dose
4  5.8      Short  0.5   Low Dose
5  6.4      Short  0.5   Low Dose
6 10.0      Short  0.5   Low Dose

> # 4. METHOD C: TEXT TRANSFORMATION (paste)
> #
> # Scenario: Create a summary column combining supplement type and length info
> df_text <- df_clean %>%
+   mutate(
+     Tooth_Summary = paste("Supplement:", supp, "| Length:", len, "| Dose:", dose)
+   )
> print("--- Method C: Text Transformation ---")
[1] "--- Method C: Text Transformation ---"
> head(df_text$Tooth_Summary)
[1] "Supplement: VC | Length: 4.2 | Dose: 0.5" "Supplement: VC | Length: 11.5 | Dose: 0.5"
[3] "Supplement: VC | Length: 7.3 | Dose: 0.5" "Supplement: VC | Length: 5.8 | Dose: 0.5"
[5] "Supplement: VC | Length: 6.4 | Dose: 0.5" "Supplement: VC | Length: 10 | Dose: 0.5"

> # 5. ALL TOGETHER (The Standard workFlow)
> #
> final_dataset <- df_clean %>%
+   mutate(
+     len_per_dose = len / dose,
+     Len_Category = ifelse(len > 20, "Long Tooth", "Short Tooth"),
+     Dose_Level = ifelse(dose >= 1, "High Dose", "Low Dose"),
+     Tooth_Report = paste0("Supplement: ", supp, ", Length: ", len, ", Dose: ", dose)
+   )

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```

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SHETH L.U.J. & SIR M.V. COLLEGE OF SCIENCE

SUBJECT - Data Analysis SAS / SPSS / R

```
RStudio
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Source
Console Terminal
R - R4.5.2 - /
1 4.2 VC 0.5 8.4 Short Tooth Low Dose
4 5.8 Short 0.5 Low Dose
5 6.4 Short 0.5 Low Dose
6 10.0 Short 0.5 Low Dose
> #
> # 4. METHOD C: TEXT TRANSFORMATION (paste)
> #
> Scenario: Create a summary column combining supplement type and length info
> df_text <- df_clean %>%
+   mutate(
+     Tooth_Summary = paste("Supplement:", supp, "| Length:", len, "| Dose:", dose)
+   )
> print("--- Method C: Text Transformation ---")
[1] "--- Method C: Text Transformation ---"
> head(df_text$Tooth_Summary)
[1] "Supplement: VC | Length: 4.2 | Dose: 0.5" "Supplement: VC | Length: 11.5 | Dose: 0.5"
[3] "Supplement: VC | Length: 7.3 | Dose: 0.5" "Supplement: VC | Length: 5.8 | Dose: 0.5"
[5] "Supplement: VC | Length: 6.4 | Dose: 0.5" "Supplement: VC | Length: 10 | Dose: 0.5"
> #
> # 5. ALL TOGETHER (The Standard workflow)
> #
> final_dataset <- df_clean %>%
+   mutate(
+     len_per_dose = len / dose,
+     len_category = ifelse(len > 20, "Long Tooth", "Short Tooth"),
+     dose_level = ifelse(dose >= 1, "High Dose", "Low Dose"),
+     Tooth_Report = paste0("Supplement: ", supp, ", Length: ", len, ", Dose: ", dose)
+   )
> print("--- Final Combined Dataset ---")
[1] "--- Final Combined Dataset ---"
> head(final_dataset)
  len supp dose len_per_dose len_category dose_level Tooth_Report
1  4.2  VC  0.5      8.4 Short Tooth Low Dose Supplement: VC, Length: 4.2, Dose: 0.5
2 11.5  VC  0.5     23.0 Short Tooth Low Dose Supplement: VC, Length: 11.5, Dose: 0.5
3  7.3  VC  0.5     14.6 Short Tooth Low Dose Supplement: VC, Length: 7.3, Dose: 0.5
4  5.8  VC  0.5     11.6 Short Tooth Low Dose Supplement: VC, Length: 5.8, Dose: 0.5
5  6.4  VC  0.5     12.8 Short Tooth Low Dose Supplement: VC, Length: 6.4, Dose: 0.5
6 10.0  VC  0.5     20.0 Short Tooth Low Dose Supplement: VC, Length: 10, Dose: 0.5
>
>
> |

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